7

Between Property (K) *and the Countable Chain Condition*

We have seen in Chapter 6 that the c.c.c. property (for compact spaces) branches into two stronger and logically independent properties: the property of calibre ω^+ and the existence of a strictly positive measure. These two properties have in fact a common denominator, stronger than c.c.c.: Knaster's property (K) (indeed, property K_n for $2 \le n < \omega$). Between the c.c.c. and property (K) there is a significant qualitative difference: the c.c.c. cannot be proved in ZFC to be productive, but property (K) is productive (Theorem 2.2(a)).

In this chapter we are concerned with differentiating property (K) from c.c.c., and with properties that lie between the two. The principal results, both assuming the continuum hypothesis and using combinatorial methods, are these:

(a) There is a c.c.c. space X such that $X \times X$ is not a c.c.c. space (Theorem 7.13, due to R. Laver and F. Galvin); and

(b) there is a productively c.c.c. space that does not have property (K) (Theorem 7.9, due to K. Kunen).

In the Notes to this chapter we describe the effect of Martin's axiom on the countable chain properties.

Kunen's example

7.1 *Lemma.* Let α be an infinite cardinal and X a space such that $S(X) \le \mathrm{cf}(\alpha)$. If $\{V_\xi : \xi < \alpha\}$ is a set of non-empty open subsets of X such that $V_{\xi'} \subset V_\xi$ for $\xi < \xi' < \alpha$, then $\{\mathrm{cl}\, V_\xi : \xi < \alpha\}$ stabilizes.

Proof. We set $W_\xi = \mathrm{int\,cl}\, V_\xi$ for $\xi < \alpha$. Since $V_\xi \subset W_\xi$, we have

$$\mathrm{cl}\, V_\xi \subset \mathrm{cl}\, W_\xi = \mathrm{cl\,int\,cl}\, V_\xi \subset \mathrm{cl\,cl}\, V_\xi = \mathrm{cl}\, V_\xi$$

and hence

$$\text{cl}\, V_\xi = \text{cl}\, W_\xi \quad \text{for } \xi < \alpha;$$

thus it is enough to show that $\{\text{cl}\, W_\xi : \xi < \alpha\}$ stabilizes.

If $\{\text{cl}\, W_\xi : \xi < \alpha\}$ does not stabilize, there is a set $\{\xi(\sigma) : \sigma < \text{cf}(\alpha)\}$ such that

$$\xi(\sigma') < \xi(\sigma) < \alpha \text{ for } \sigma' < \sigma < \text{cf}(\alpha) \text{ and } \text{cl}\, W_{\xi(\sigma)} \backslash \text{cl}\, W_{\xi(\sigma+1)} \neq \varnothing.$$

It follows that $W_{\xi(\sigma)} \backslash \text{cl}\, W_{\xi(\sigma+1)} \neq \varnothing$, since otherwise from

$$W_{\xi(\sigma)} \subset \text{cl}\, W_{\xi(\sigma+1)} \subset \text{cl}\, W_{\xi(\sigma)}$$

we have the contradiction $\text{cl}\, W_{\xi(\sigma)} = \text{cl}\, W_{\xi(\sigma+1)}$. We note finally that for $\sigma' < \sigma < \text{cf}(\alpha)$ we have $\xi(\sigma'+1) \leq \xi(\sigma)$ and hence

$$\begin{aligned}(W_{\xi(\sigma')} \backslash \text{cl}\, W_{\xi(\sigma'+1)}) \cap (W_{\xi(\sigma)} \backslash \text{cl}\, W_{\xi(\sigma+1)}) &\subset W_{\xi(\sigma)} \backslash \text{cl}\, W_{\xi(\sigma'+1)} \\ &\subset W_{\xi(\sigma'+1)} \backslash \text{cl}\, W_{\xi(\sigma'+1)} = \varnothing;\end{aligned}$$

thus $\{W_{\xi(\sigma)} \backslash \text{cl}\, W_{\xi(\sigma+1)} : \sigma < \text{cf}(\alpha)\}$ is a (faithfully indexed) cellular family in X, contradicting the assumption $S(X) \leq \text{cf}\,(\alpha)$.

We recall from Chapter 2 that for α an infinite cardinal and ρ an ordinal such that $\rho \leq \alpha$, a space X has precalibre (α, ρ) if for every set $\{U_\xi : \xi < \alpha\}$ of non-empty, open subsets of X there is $A \subset \alpha$ with type $A = \rho$ such that $\{U_\xi : \xi \in A\}$ has the finite intersection property.

7.2 *Theorem.* Let X be a c.c.c. space and ρ an ordinal such that $\rho < \omega^+$. Then X has precalibre (ω^+, ρ).

Proof. It follows from Corollary 2.22 that it is enough to show that the Gleason space $G(X)$ of X has precalibre (ω^+, ρ); we assume without loss of generality in what follows that X is the (compact, Hausdorff) space $G(X)$.

Let $\{U_\eta : \eta < \omega^+\}$ be a set of non-empty, open subsets of X, and set

$$V_\xi = \bigcup_{\xi \leq \eta < \omega^+} U_\eta \quad \text{for } \xi < \omega^+.$$

Since $V_{\xi'} \subset V_\xi$ for $\xi < \xi' < \omega^+$ and X is a c.c.c. space, it follows from (the case $\alpha = \omega^+$ of) Lemma 7.1 that there is $\bar{\xi} < \omega^+$ such that

$$\text{cl}\, V_\xi = \text{cl}\, V_{\bar{\xi}} \quad \text{for } \bar{\xi} \leq \xi < \omega^+,$$

i.e., such that V_ξ is dense in $\text{cl}\, V_{\bar{\xi}}$ for $\bar{\xi} \leq \xi < \omega^+$.

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We claim that for $\bar{\xi} < \xi < \omega^+$ there is $f(\xi) < \omega^+$ such that

$$\cup\{U_\eta : \xi \le \eta < f(\xi)\}$$

is dense in cl $V_{\bar{\xi}}$. We set

$$\mathscr{W} = \{W \in \mathscr{T}^*(X) : \text{there is } \eta \ge \xi \text{ with } W \subset U_\eta\},$$

and we choose a maximal (faithfully indexed) cellular subfamily $\{W_i : i \in I\}$ of $\mathscr{W}$. We have $|I| \le \omega$ (since X is a c.c.c. space), and from the maximality condition it follows that $\cup_{i \in I} W_i$ is dense in V_ξ and hence in cl V_ξ. For $i \in I$ there is $\eta(i)$ such that $\xi \le \eta(i) < \omega^+$ and $W_i \subset U_{\eta(i)}$; we set

$$f(\xi) = \sup\{\eta(i) : i \in I\} + 1.$$

The claim is proved.

We define $g : \omega^+ \to \omega^+$ by the rule

$$\begin{aligned} &g(0) = \bar{\xi}, \\ &g(\xi + 1) = f(g(\xi)) \quad \text{for } \xi < \omega^+, \text{ and} \\ &g(\xi) = \sup\{g(\xi') : \xi' < \xi\} \end{aligned}$$

for non-zero limit ordinals $\xi < \omega^+$, and for $\xi < \rho$ we set $X_\xi = \cup\{U_\eta : g(\xi) \le \eta < g(\xi + 1)\}$. Then $\{X_\xi : \xi < \rho\}$ is a set of dense, open subspaces of the compact, Hausdorff space cl $V_{\bar{\xi}}$. Since ρ is a countable ordinal, it follows from the Baire category theorem that there is $p \in \cap_{\xi < \rho} X_\xi$. For $\xi < \rho$ there is $\eta(\xi)$ such that

$$\begin{aligned} &g(\xi) \le \eta(\xi) < g(\xi + 1) \quad \text{and} \\ &p \in U_{\eta(\xi)}. \end{aligned}$$

Since the function $\xi \to \eta(\xi)$ is an ordered-set isomorphism from ρ into ω^+ we have

$$\text{type}\{\eta(\xi) : \xi < \rho\} = \rho,$$

and since $p \in \cap_{\xi < \rho} U_{\eta(\xi)}$ the set $\{U_{\eta(\xi)} : \xi < \rho\}$ has the finite intersection property.

The proof is complete.

7.3 *Problem.* Let $\kappa > \omega^+$. Is there a c.c.c. space X such that X does not have precalibre (κ, ω^+)?

Definition. A space is *productively c.c.c.* if its product with every c.c.c. space is a c.c.c. space.

Definition. A space X has *property* P if for every set $\{U_\xi : \xi < \omega^+\}$ of non-empty, open subsets of X there are $A \subset \omega^+$ with $|A| = \omega^+$ and an ordinal number ρ with $2 \leq \rho < \omega^+$ such that:

if $B \subset A$ and type $B = \rho$ then there are $\xi, \xi' \in B$ with $\xi \neq \xi'$ such that $U_\xi \cap U_{\xi'} \neq \varnothing$.

We remark that if X and Y are spaces and $X \equiv Y$, then X has property P if and only if Y has property P. (The proof is similar to the proof of Corollary 2.18(c); we omit the details.) It follows from Theorem 2.20 that a space X has property P if and only if its Gleason space $G(X)$ has property P.

7.4 *Theorem.* Let X be a space.

(a) If X has property (K) then X has property P.

(b) If X has property P then X is productively c.c.c.

Proof. (a) This statement is immediate from the definitions; one may take for ρ any ordinal such that $2 \leq \rho < \omega^+$.

(b) Let X have property P, let Y be a c.c.c. space and let $\{U_\xi \times V_\xi : \xi < \omega^+\}$ be a set of non-empty, open subsets of $X \times Y$. We show there are $\xi, \xi' < \omega^+$ with $\xi \neq \xi'$ such that

$$(U_\xi \times V_\xi) \cap (U_{\xi'} \times V_{\xi'}) \neq \varnothing.$$

There are $A \subset \omega^+$ with $|A| = \omega^+$ and an ordinal ρ with $2 \leq \rho < \omega^+$ such that if $B \subset A$ with type $B = \rho$ then there are $\xi, \xi' \in B$ with $\xi \neq \xi'$ such that $U_\xi \cap U_{\xi'} \neq \varnothing$. It follows from Theorem 7.2 that Y has precalibre (ω^+, ρ); we choose $\bar{B} \subset A$ with type $\bar{B} = \rho$ such that $\{V_\xi : \xi \in \bar{B}\}$ has the finite intersection property.

There are $\bar{\xi}, \bar{\xi}' \in \bar{B}$ with $\bar{\xi} \neq \bar{\xi}'$ such that $U_{\bar{\xi}} \cap U_{\bar{\xi}'} \neq \varnothing$. It is clear that

$$(U_{\bar{\xi}} \times V_{\bar{\xi}}) \cap (U_{\bar{\xi}'} \times V_{\bar{\xi}'}) \neq \varnothing,$$

as required.

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7.5 *Corollary.* Let X be a space such that either X has calibre ω^+ or X has a strictly positive measure. Then X is productively c.c.c.

Proof. It is clear that if X has calibre ω^+ then X has property (K) (cf. Theorem 2.1(a)); it follows from Lemmas 6.2 and 6.22 (or from Corollary 6.17) that if X has a strictly positive measure then X has property (K). Thus the required statement follows from Theorem 7.4.

7.6 *Problems.* (a) (Galvin) Assume the continuum hypothesis. Is there a productively c.c.c. space that does not satisfy property P?

(b) (Argyros) Is there a (c.c.c.) space X that can serve as a 'test space' for productively c.c.c. spaces in the following sense: if Y is a space and $X \times Y$ is a c.c.c.space, then Y is productively c.c.c.

It is conjectured that (a) will be answered in the affirmative and (b) in the negative.

7.7 *Lemma.* Assume the continuum hypothesis. There is a family $\{S_\eta : \eta < \omega^+\}$ of subsets of ω^+ such that

(a) $S_\eta \subset \eta \quad$ for $\eta < \omega^+$;

(b) $|S_\eta \cap S_\zeta| < \omega \quad$ for $\eta < \zeta < \omega^+$; and

(c) for all $X \subset \omega^+$, either

(i) there is finite $F \subset \omega^+$ such that $X \subset \cup_{\eta \in F} S_\eta$, or

(ii) there is $\eta < \omega^+$ such that $X \cap S_\zeta \neq \varnothing$ for $\eta < \zeta < \omega^+$.

Proof. From the continuum hypothesis we have $|\{A \subset \omega^+ : |A| = \omega\}| = \omega^+$. Let $\{A_\eta : \eta < \omega^+\}$ be a well-ordering of this set of countable subsets of ω^+. We define $\{S_\eta : \eta < \omega^+\}$ satisfying conditions (a), (b) and (d) defined as follows:

(d) if $\xi < \eta < \omega^+$ with $A \subset \eta$, and if there is no finite $F \subset \eta$ with $A_\xi \subset \cup_{\zeta \in F} S_\zeta$, then $A_\xi \cap S_\eta \neq \varnothing$.

We proceed by recursion.

We set $S_0 = \varnothing$.

If $\eta < \omega^+$, and if S_ζ has been defined for $\zeta < \eta$, we enumerate $\{S_\zeta : \zeta < \eta\}$ as $\{S^{(n)} : n < \omega\}$ and we enumerate

$$\{A_\xi : A_\xi \subset \eta, \xi < \eta, \text{ there is no finite } F \subset \eta \text{ with } A_\xi \subset \cup_{\zeta \in F} S_\zeta\}$$

as $\{A^{(n)} : n < \omega\}$; we note that

$$A^{(n)} \not\subset \bigcup_{k \le n} S^{(k)} \quad \text{for } n < \omega,$$

we choose

$$s_n \in A^{(n)} \backslash \bigcup_{k \le n} S^{(k)} \quad \text{for } n < \omega,$$

and we set

$$S_\eta = \{s_n : n < \omega\}.$$

The definition of the family $\{S_\eta : \eta < \omega^+\}$ is complete. It is clear that conditions (a) and (d) are satisfied; we note further that if $\zeta < \eta$ and $S_\zeta = S^{(k)}$ (in the enumeration used to define S_η) then

$$S_\eta \cap S_\zeta = S_\eta \cap S^{(k)} \subset \{s_n : n < k\}$$

and hence $|S_\eta \cap S_\zeta| < \omega$.

It remains to verify (c). If X is a countable subset of ω^+ there is $\xi < \omega^+$ such that $X = A_\xi$ and condition (c) for X follows from (d). In what follows we assume $|X| = \omega^+$ and we show there is $Y \subset X$ with $|Y| = \omega$ such that Y satisfies (c) (ii).

Let $Y_0 \subset X$ with $|Y_0| = \omega$. If (c)(i) fails for Y_0 we set $Y = Y_0$; and if (c) (i) holds for Y_0 we choose $\eta_0 < \omega^+$ such that $|Y_0 \cap S_{\eta_0}| = \omega$ and we choose $Y_1 \subset X \backslash \cup_{\eta \le \eta_0} S_\eta$ such that $|Y_1| = \omega$. If (c)(i) fails for Y_1 set $Y = Y_1$; and if (c)(i) holds for Y_1 we choose $\eta_1 < \omega^+$ with $\eta_1 > \eta_0$ such that $|Y_1 \cap S_{\eta_1}| = \omega$.

We continue recursively. If there is (minimal) $n < \omega$ such that (c)(i) fails for Y_n we set $Y = Y_n$ and the proof is complete. Otherwise Y_n is defined for $n < \omega$ and we set

$$Y = \bigcup_{n < \omega} Y_n;$$

then $|Y| = \omega$, and since $|\{\eta < \omega^+ : |Y \cap S_\eta| = \omega\}| = \omega$ there is no finite $F \subset \omega^+$ such that $Y \subset \cup_{\eta \in F} S_\eta$. There is $\bar{\xi} < \omega^+$ such that $Y = A_{\bar{\xi}}$ and there is $\bar{\eta} < \omega^+$ such that $A_{\bar{\xi}} \subset \bar{\eta}$. It follows from (d) that if $\bar{\eta} < \zeta < \omega^+$ then

$$Y \cap S_\zeta = A_{\bar{\xi}} \cap S_\zeta \neq \varnothing,$$

so that Y satisfies (c)(ii), as required.

The proof is complete.

In passing we prove the following arrow relation; compare Theorem 1.7.

7.8 *Corollary*. Assume the continuum hypothesis. Then $\omega^+ \nrightarrow (\omega^+, \omega+2)$.

Proof. Let $\{S_\eta : \eta < \omega^+\}$ be a family of subsets of ω^+ satisfying conditions (a), (b) and (c) of Lemma 7.7, and set

$$P_1 = \{A \in [\omega^+]^2 : \min A \in S_{\max A}\} \text{ and}$$
$$P_0 = [\omega^+]^2 \backslash P_1.$$

Let $A \subset \omega^+$ with $|A| = \omega^+$. Since condition (c)(i) fails for A there is $\eta < \omega^+$ such that $A \cap S_\zeta \neq \varnothing$ for $\eta < \zeta < \omega^+$. We choose $\zeta \in A$ such that $\eta < \zeta < \omega^+$ and then $\xi \in A \cap S_\zeta$; we have $\xi < \zeta$ from condition (a) and hence $\{\xi, \zeta\} \in P_1$. It follows that $[A]^2 \not\subset P_0$.

Now let $B \subset \omega^+$ with type $B = \omega + 2$, let η and ζ be the last two elements of B, and set

$$C = B \backslash \{\eta, \zeta\}.$$

If $[B]^2 \subset P_1$ then for $\xi \in C$ we have ($\xi < \eta$ and) $\xi \in S_\eta$; similarly $\xi \in S_\zeta$. Thus if $[B]^2 \subset P_1$ then $S_\eta \cap S_\zeta$ contains the infinite set C, contrary to condition (b). If follows that $[B]^2 \not\subset P_1$.

The proof is complete.

7.9 *Theorem*. Assume the continuum hypothesis. There is an extremally disconnected, compact Hausdorff space X such that

(i) X does not have property (K), and

(ii) X has property P (hence, X is productively c.c.c.).

Proof. It follows from Corollary 2.22(b) and the remark preceding Theorem 7.4 that it is enough to find a space X with property P that does not have property (K); for then the Gleason space of X will have the required properties.

It follows from Lemma 7.7 that there is a family $\{S_\eta : \eta < \omega^+\}$ of subsets of ω^+ such that

(a) $S_\eta \subset \eta$ for $\eta < \omega^+$,

(b) $|S_\eta \cap S_{\eta'}| < \omega$ for $\eta < \eta' < \omega^+$, and

(c) if $X \subset \omega^+$ then either

  (1) there is finite $F \subset \omega^+$ such that $X \subset \cup_{\eta \in F} S_\eta$, or

  (2) there is $\bar{\eta} < \omega^+$ such that $X \cap S_\eta \neq \varnothing$ for $\bar{\eta} \leq \eta < \omega^+$.

Let X as a set be equal to $\{0,1\}^{\omega^+}$, set $V_\eta = \{x \in X : x | S_\eta \equiv 0, x_\eta = 1\}$ for $\eta < \omega^+$, and give X the topology determined by the subbase $\{V_\eta : \eta < \omega^+\}$.

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We verify that the space X satisfies (i) and (ii).

(i) We consider the family $\{V_\eta : \eta < \omega^+\}$ and we let $A \subset \omega^+$ with $|A| = \omega^+$. There is no finite $F \subset \omega^+$ with $A \subset \cup_{\eta \in F} S_\eta$; hence from condition (c) there is $\bar{\eta} < \omega^+$ such that

$$A \cap S_\eta \neq \varnothing \quad \text{for } \bar{\eta} \leq \eta < \omega^+.$$

There is $\eta \in A$ such that $\bar{n} \leq \eta$ and hence there is $\eta' \in A \cap S_\eta$. It is clear that

$$V_\eta \cap V_{\eta'} = \varnothing,$$

as required.

(ii) Let $\{U_\xi : \xi < \omega^+\}$ be a family of non-empty, basic open subsets of X; for $\xi < \omega^+$ there is a finite subset $F(\xi)$ of ω^+ such that

$$U_\xi = \bigcap_{\eta \in F(\xi)} V_\eta.$$

For $\xi < \alpha$ there is $X \in U_\xi$ and since $x|F(\xi) \equiv 1$ and $x|(\cup_{\eta \in F(\xi)} S_\eta) \equiv 0$ we have

$$\Big(\bigcup_{\eta \in F(\xi)} S_\eta\Big) \cap F(\xi) = \varnothing.$$

We assume without loss of generality, using the Erdős – Rado theorem for quasi-disjoint families (Theorem 1.4), that there is a set J such that

$$F(\xi) \cap F(\xi') = J \quad \text{for } \xi < \xi' < \omega^+;$$

we set

$$G(\xi) = F(\xi) \backslash J \quad \text{for } \xi < \omega^+.$$

We assume further without loss of generality that there is $n < \omega$ such that

$$|G(\xi)| = n \quad \text{for all } \xi < \omega^+.$$

If $n = 0$ we have

$$F(\xi) = F(\xi') \quad \text{for } \xi < \xi' < \omega^+$$

and the requirement of property P is satisfied with $A = \omega^+$, $\rho = 2$. We assume in what follows that $n > 0$ and for $\xi < \omega^+$ we enumerate $G(\xi)$ in its order inherited from ω^+:

$$G(\xi) = \{g_\xi(1) < g_\xi(2) < \ldots < g_\xi(n)\}.$$

We note that since $\{G(\xi) : \xi < \omega^+\}$ is a family of pairwise disjoint, finite subsets of ω^+ there is $A \subset \omega^+$ with $|A| = \omega^+$ such that if $\xi, \xi' \in A$ with $\xi < \xi'$ then $G(\xi) < G(\xi')$ (in the sense that if $\zeta \in G(\xi)$ and $\zeta' \in G(\xi')$ then $\zeta < \zeta'$). (Indeed set $\xi(0) = 0$. If $\zeta < \omega^+$ and $\xi(\eta)$ has been defined for all $\eta < \zeta$, note that

$$\sup \left(\bigcup_{\eta < \zeta} G(\xi(\eta)) < \omega^+ \right.$$

and choose for $\xi(\zeta)$ any ordinal $\xi < \omega^+$ such that

$$\sup \left(\bigcup_{\eta < \zeta} G(\xi(\eta)) \right) < g_\xi(1).$$

This defines $\xi(\zeta)$ for all $\zeta < \omega^+$; set $A = \{\xi(\zeta) : \zeta < \omega^+\}$.)

We claim that the requirement of property P is satisfied with the set A defined above and with $\rho = \omega + n + 1$. Indeed we claim that if $B \subset A$ with type $B = \omega + n + 1$, say

$$B = \{\xi_0 < \xi_1 < \ldots < \xi_k < \ldots \xi_\omega < \xi_{\omega+1} < \ldots < \xi_{\omega+n}\},$$

then there are k, i with $k < \omega$ and $0 \leq i \leq n$ such that

$$F(\xi_k) \cup F(\xi_{\omega+i}) \cap (\cup \{S_\eta : \eta \in F(\xi_k) \cup F(\xi_{\omega+i})\}) = \varnothing$$

(and hence $V_{\xi_k} \cap V_{\xi_{\omega+i}} \neq \varnothing$). If the claim fails then for $k < \omega$, $0 \leq i \leq n$ there is $\eta \in F(\xi_k) \cup F(\xi_{\omega+i})$ such that

$$(F(\xi_k) \cup F(\xi_{\omega+i})) \cap S_\eta \neq \varnothing.$$

Since $F(\xi_k) \cap F(\xi_{\omega+i}) = J$ and

$$F(\xi_k) \cap \left(\bigcup_{\eta \in F(\xi_k)} S_\eta \right) = F(\xi_{\omega+i}) \cap \left(\bigcup_{\eta \in F(\xi_{\omega+i})} S_\eta \right) = \varnothing$$

it is clear that in fact $\eta \in G(\xi_k) \cup G(\xi_{\omega+i})$ and

$$(G(\xi_k) \cup G(\xi_{\omega+i})) \cap S_\eta \neq \varnothing.$$

Since for $\eta \in G(\xi_k)$ we have

$$G(\xi_{\omega+i}) \cap S_\eta \subset G(\xi_{\omega+i}) \cap \eta = \varnothing,$$

it follows that for $k < \omega, 0 \leq i \leq n$ there is $\eta \in G(\xi_{\omega+i})$ such that $G(\xi_k) \cap S_\eta \neq \varnothing$. We define functions

$$\psi, \varphi : \omega \times \{0, 1. \ldots, n\} \to \{1, 2, \ldots, n\}$$

by the rule

$$\eta = g_{\xi_{\omega+i}}(\psi(k, i)) \in G(\xi_{\omega+i}) \quad \text{and } g_{\xi_k}(\varphi(k, i)) \in G(\xi_k) \cap S_\eta.$$

We note that there is an infinite subset K of ω such that for each i

with $0 \leq i \leq n$ the two functions

$$\psi|K \times \{i\}, \quad \varphi|K \times \{i\}$$

are constant functions. (Indeed there is infinite $K_0 \subset \omega$ such that $\psi|K_0 \times \{0\}$, $\varphi|K_0 \times \{0\}$ are constant. If $m < n$ and infinite K_m has been defined such that $\psi|K_m \times \{i\}$, $\varphi|K_m \times \{i\}$ are constant for $0 \leq i \leq m$, there is infinite $K_{m+1} \subset K_m$ such that $\psi|K_{m+1} \times \{m+1\}$, $\varphi|K_{m+1} \times \{m+1\}$ are constant. This defines K_m for $m \leq n$. The set $K = K_n$ is as required.)

Since $|\{0, 1, \ldots, n\}| = n + 1$ and $|\{1, 2, \ldots, n\}| = n$ there are $\bar{i}, \bar{j}$ such that $0 \leq \bar{i} < \bar{j} \leq n$ and the constant values of $\varphi|K \times \{\bar{i}\}$ and $\varphi|K \times \{\bar{j}\}$ are equal; that is, there is r such that $1 \leq r \leq n$ and

$$\varphi(k, \bar{i}) = \varphi(k, \bar{j}) = r \quad \text{for all } k \in K.$$

Let

$$\psi(k, \bar{i}) = m_0 \quad \text{and } \psi(k, \bar{j}) = m_1 \quad \text{for all } k \in k,$$

and set

$$\zeta(\bar{i}) = g_{\xi_{\omega+\bar{i}}}(m_0) \quad \text{and } \zeta(\bar{j}) = g_{\xi_{\omega+\bar{j}}}(m_1).$$

Since $\zeta(\bar{i}) \in G(\xi_{\omega+\bar{i}})$ and $\zeta(\bar{j}) \in G(\xi_{\omega+\bar{j}})$ and

$$G(\xi_{\omega+\bar{i}}) \cap G(\xi_{\omega+\bar{j}}) = \varnothing$$

we have $\zeta(\bar{i}) \neq \zeta(\bar{j})$.

Now for $k \in K$ we have

$$\zeta(\bar{i}) = g_{\xi_{\omega+\bar{i}}}(\psi(k, \bar{i}))$$

and hence

$$g_{\xi_k}(r) = g_{\xi_k}(\varphi(k, \bar{i})) \in G(\xi_k) \cap S_{\zeta(\bar{i})} \subset S_{\zeta(\bar{i})};$$

similarly for $k \in K$ we have

$$g_{\xi_k}(r) \in S_{\zeta(\bar{j})}.$$

Since $g_{\xi_k}(r) \in G(\xi_k)$ and

$$G(\xi_k) \cap G(\xi_{k'}) = \varnothing \quad \text{for } k, k' \in K, k \neq k'$$

we have

$$g_{\xi_k}(r) \neq g_{\xi_{k'}}(r) \quad \text{for } k, k' \in K, k \neq k'.$$

It follows that

$$|S_{\zeta(\bar{i})} \cap S_{\zeta(\bar{j})}| = \omega,$$

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contrary to condition (b) for the family $\{S_\eta : \eta < \omega^+\}$. This contradiction completes the proof.

7.10 *Remark and Problem.* For $\beta \geq \omega$, we say that a space X has *property* P_β if for every set $\{U_\xi : \xi < \beta\}$ of non-empty, open subsets of X there are $A \subset \beta$ with $|A| = \beta$ and an ordinal number ρ with $2 \leq \rho < \beta$ such that:

if $B \subset A$ and type $B = \rho$ then there are $\xi, \xi' \in B$ with $\xi \neq \xi'$ such that $U_\xi \cap U_{\xi'} \neq \varnothing$.

It is clear that the method of Theorem 7.9 proves the following statement: If α is a regular cardinal and $\alpha^+ = 2^\alpha$ then there is an extremally disconnected, compact Hausdorff space X such that

X has property P_{α^+} and

X does not have property $\mathrm{K}_{\alpha^+,2}$.

It is clear then that $S(X) \leq \alpha^+$, but we do not know if X has the stronger productive property: $S(X \times Y) \leq \alpha^+$ for every space Y with $S(Y) \leq \alpha^+$. This last must be left as a question.

The Laver–Galvin example

7.11 *Lemma.* Let $\alpha \geq \omega$ and A a set, and for $\xi < \alpha$ let $\{F^i_\xi : i \in I_\xi\}$ be a set of pairwise disjoint, finite subsets of A with $|I_\xi| = \alpha^\eta$. Then there are two subsets A_0, A_1 of A such that

$$A_0 \cap A_1 = \varnothing, \text{ and}$$
$$|\{i \in I_\xi : F^i_\xi \subset A_\varepsilon\}| = \alpha \text{ for } \xi < \alpha \text{ and } \varepsilon = 0, 1.$$

Proof. Without loss of generality we assume $I_\xi = \alpha$ for $\xi < \alpha$. We note that it follows easily from the assumptions that if B is a set with $|B| < \alpha$, if φ and ψ are functions from B to α and if $\xi < \alpha$, then there is $i < \alpha$ such that

$$(1) \quad F^i_\xi \cap (\cup_{\zeta \in B} F^{\psi(\zeta)}_{\varphi(\zeta)}) = \varnothing.$$

We define by recursion a function

$$\tau : \{\langle \xi, \eta \rangle : \xi \leq \eta < \alpha\} \to \alpha$$

such that

$$F^{\tau(\xi,\eta)}_\xi \cap F^{\tau(\xi',\eta)}_{\xi'} = \varnothing \text{ for } \langle \xi, \eta \rangle \neq \langle \xi', \eta' \rangle.$$

We set $\tau(0,0)=0$. If $\xi \le \eta < \alpha$ and if τ has been defined on the set

$$\{\langle \xi', \eta' \rangle : \xi' \le \eta' < \eta\} \cup \{\langle \xi', \eta \rangle : \xi' < \eta\},$$

we define $\tau(\xi,\eta)$, using (1), so that

$$F_\xi^{\tau(\xi,\eta)} \cap (\cup_{\xi' \le \eta' < \eta} F_{\xi'}^{\tau(\xi',\eta')} \cup \cup_{\xi' < \xi} F_{\xi'}^{\tau(\xi',\eta)}) = \varnothing.$$

We set

$$J_\xi = \{\tau(\xi,\eta) : \xi \le \eta < \alpha\} \text{ for } \xi < \alpha,$$

we choose sets $J_{\xi,0}$ and $J_{\xi,1}$ such that

$$J_{\xi,0} \cup J_{\xi,1} = J_\xi,$$
$$J_{\xi,0} \cap J_{\xi,1} = \varnothing, \text{ and}$$
$$|J_{\xi,0}| = |J_{\xi,1}| = \alpha,$$

and we set

$$A_\varepsilon = \cup \{F_\xi^\zeta : \xi < \alpha, \zeta \in J_{\xi,\varepsilon}\} \text{ for } \varepsilon = 0, 1.$$

The proof is complete.

Notation. For sets A, B we set

$$A \otimes B = \{\{a,b\} : a \in A, b \in B\}.$$

7.12 *Lemma.* Let $\alpha \ge \omega$ and $\alpha^+ = 2^\alpha$. There are two families $\{K_0(\eta) : \eta < \alpha^+\}, \{K_1(\eta) : \eta < \alpha^+\}$ such that (with $K_\varepsilon = \cup_{\eta < \alpha^+} (K_\varepsilon(\eta) \times \{\eta\})$ and $\varepsilon = 0, 1$):

(i) $K_0(\eta) \cup K_1(\eta) = \eta$ for $\eta < \alpha^+$, and

(ii) if $\mathscr{F} = \{F^i : i < \alpha\}$ is a set of finite subsets of α^+ such that

$$F^i \cap F^{i'} = \varnothing \text{ for } i < i' < \alpha,$$

then there is $\zeta < \alpha^+$ such that if $\zeta < \eta < \alpha^+$ and $\cup \mathscr{F} \subset \eta$ and $X \subset \eta$ with $|X| < \omega$ and $\varepsilon \in \{0, 1\}$ and

$$|\{i < \alpha : (F^i \otimes X) \cap K_\varepsilon = \varnothing\}| = \alpha,$$

then $|\{i < \alpha : F^i \otimes (X \cup \{\eta\}) \cap K_\varepsilon = \varnothing\}| = \alpha.$

Proof. For $\eta < \alpha^+$ we define sets $L_\varepsilon(\eta)$ for $\varepsilon = 0, 1$ such that

$L_\varepsilon(\eta) \subset \eta$,

$L_0(\eta) \cap L_1(\eta) = \varnothing$, and

if $\mathscr{F} = \{F^i : i < \alpha\}$ is a set of finite subsets of α such that

$F^i \cap F^{i'} = \varnothing$ for $i < i' < \alpha$

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then there is $\zeta < \alpha^+$ such that if $\zeta < \eta < \alpha^+$ and $\cup \mathscr{F} \subset \eta$ and $X \subset \eta$ with $|X| < \eta$ and $\varepsilon \in \{0, 1\}$ and

$$|\{i < a : F^i \otimes X \subset \cup_{\eta' < \eta}(L_\varepsilon(\eta') \otimes \{\eta'\})\}| = \alpha,$$

then $|\{i < \alpha : F^i \otimes (X \cup \{\eta\}) \subset \cup_{\eta' < \eta}(L_\varepsilon(\eta') \otimes \{\eta'\})\}| = \alpha.$

We proceed by recursion.

We note that the set of all families $\mathscr{F} = \{F^i : i < \alpha\}$ of finite subsets of α such that

$$F^i \cap F^{i'} = \varnothing \text{ for } i < i' < \alpha$$

has cardinality $|\mathscr{P}_\omega(\alpha^+)|^\alpha = (\alpha^+)^\alpha = (2^\alpha)^\alpha = \alpha^+$. Let $\{\mathscr{F}_\zeta : \zeta < \alpha^+\}$ be a well-ordering of this set.

We set $L_\varepsilon(\eta) = \varnothing$ for $\varepsilon = 0, 1, \eta \le \alpha$.

Let $\alpha < \eta < \alpha^+$, suppose that $L_\varepsilon(\eta')$ has been defined for $\varepsilon = 0, 1$ and $\eta' < \eta$, and let

$$\{(\varepsilon_\xi, \zeta_\xi, X_\xi) : \xi < \alpha\}$$

be a well-ordering of the set of ordered triples (ε, ζ, X) such that

$$\varepsilon \in \{0, 1\}, \zeta < \eta, \cup \mathscr{F}_\zeta \subset \eta, X \subset \eta, |X| < \omega \text{ and}$$
$$|\{i < \alpha : F^i_\zeta \otimes X \subset \cup_{\eta' < \eta}(L_\varepsilon(\eta') \otimes \{\eta'\})\}| = \alpha.$$

We set

$$I_\xi = \{i < \alpha : F^i_{\zeta_\xi} \otimes X_\xi \subset \cup_{\eta' < \eta}(L_{\varepsilon_\xi}(\eta') \otimes \{\eta'\})\} \text{ for } \xi < \alpha$$

and we apply Lemma 7.11 with A and F_ξ replaced by η and $F^i_{\zeta_\xi}$, respectively: there are two sets $L_0(\eta), L_1(\eta)$ such that

$$L_0(\eta), L_1(\eta) \subset \eta,$$
$$L_0(\eta) \cap L_1(\eta) = \varnothing, \text{ and}$$
$$|\{i \in I_\xi : F^i_{\zeta_\xi} \subset L_{\varepsilon_\xi}(\eta)\}| = \alpha.$$

The definition of the family $\{L_\varepsilon(\eta) : \eta < \alpha^+, \varepsilon = 0, 1\}$ is complete. We finally set

$$K_\varepsilon(\eta) = \eta \backslash L_\varepsilon(\eta) \text{ for } \varepsilon = 0, 1 \text{ and } \eta < \alpha^+.$$

It is clear that the two families $\{K_\varepsilon(\eta) : \eta < \alpha^+\}$ $(\varepsilon \in \{0, 1\})$ are as required.

7.13 *Theorem.* Let $\alpha \ge \omega$ and $\alpha^+ = 2^\alpha$. There is an extremally disconnected, compact Hausdorff space X such that

$$S(X) \leq \alpha^+ \text{ and } S(X \times X) > \alpha^+.$$

Proof. We show first that there are two spaces X_0, X_1 such that $S(X_0) \leq \alpha^+, S(X_1) \leq \alpha^+$, and $S(X_0 \times X_1) > \alpha^+$.

For $\varepsilon \in \{0, 1\}$ let $\{K_\varepsilon(\eta) : \eta < \alpha^+\}$ be the family of Lemma 7.12, let X_ε as a set be equal to $\{0, 1\}^{(\alpha^+)}$, set

$$V_\varepsilon(\eta) = \{x \in X_\varepsilon : x | K_\varepsilon(\eta) \equiv 0, x(\eta) = 1\} \text{ for } \eta < \alpha^+,$$

and give X_ε the topology determined by the subbase $\{V_\varepsilon(\eta) : \eta < \alpha^+\}$.

(i) $S(X_0 \times X_1) > \alpha^+$. We claim the (indexed) family $\{V_0(\eta) \times V_1(\eta) : \eta < \alpha^+\}$ is a cellular family in $X_0 \times X_1$. Indeed if there are η, η' with $\eta < \eta' < \alpha^+$ and

$$\langle x_0, x_1 \rangle \in (V_0(\eta) \times V_1(\eta)) \cap (V_0(\eta') \times V_1(\eta'))$$

then since $\eta \in K_0(\eta') \cup K_1(\eta') = \eta'$ there is $\varepsilon \in \{0, 1\}$ such that $\eta \in K_\varepsilon(\eta')$; we have $x_\varepsilon(\eta) = 0$ and $x_\varepsilon(\eta) = 1$, a contradiction.

(ii) $S(X_0) \leq \alpha^+, S(X_1) \leq \alpha^+$. Let $\varepsilon \in \{0, 1\}$ and let $\{U_i : i < \alpha^+\}$ be a family of non-empty, basic open subsets of X_ε; for $i < \alpha^+$ there is a finite subset G^i of α^+ such that

$$U_i = \bigcap_{\eta \in G^i} V_\varepsilon(\eta).$$

For $i < \alpha^+$ there is $x \in U_i$ and since $x | G^i \equiv 1$ and $x | \cup_{\eta \in G^i} K_\varepsilon(\eta) \equiv 0$ we have

$$\left(\bigcup_{\eta \in G^i} K_\varepsilon(\eta)\right) \cap G^i = \varnothing.$$

We assume without loss of generality, using the Erdős–Rado theorem for quasi-disjoint families (Theorem 1.4), that there is a set J such that

$$G^i \cap G^{i'} = J \text{ for } i < i' < \alpha^+;$$

and we set $F^i = G^i \backslash J$ for $i < \alpha^+$. We note that if $i' < i < \alpha^+$ and $(F^i \otimes F^{i'}) \cap K_\varepsilon = \varnothing$ then $U_i \cap U_{i'} \neq \varnothing$. Thus to prove that $\{U_i : i < \alpha^+\}$ is not a cellular family in X_ε it is enough to show that there are i', i such that ($i' < \alpha < i < \alpha^+$ and) $F^i \otimes F^{i'} \cap K_\varepsilon = \varnothing$.

Set $\mathscr{F} = \{F^i : i < \alpha\}$ and let $\zeta < \alpha^+$ be an ordinal satisfying condition (ii) of Lemma 7.12 for $\mathscr{F}$; we assume without loss of generality that $\cup \mathscr{F} \subset \zeta$. There is $\bar{\imath} < \alpha^+$ such that

$$\alpha < \bar{\imath} \text{ and } F^{\bar{\imath}} \cap (\zeta + 1) = \varnothing.$$

We note that if $\eta \in F^{\bar{\imath}}$ then $\eta > \zeta$ and $\cup_{i<\alpha} F^i \subset \eta$; further if $F^{\bar{\imath}} = \varnothing$ then the required relation $(F^{i'} \otimes F^{\bar{\imath}}) \cap K_\varepsilon = \varnothing$ is satisfied for all $i' < \alpha$. We assume therefore that $F^{\bar{\imath}} \neq \varnothing$, we let

$$\{\eta_1 < \eta_2 < \ldots < \eta_n\}$$

be the elements of $F^{\bar{\imath}}$ in the order inherited from α^+, we note that

$$\{i < \alpha : (F^i \otimes \varnothing) \cap (\cup_{\eta' < \eta_1} (K_\varepsilon(\eta') \otimes \{\eta'\})) = \varnothing\} = \alpha,$$

and we apply condition (ii) of Lemma 7.12 successively n times, with

$$\eta = \eta_1, \eta_2, \ldots, \eta_n \text{ and}$$
$$X = \varnothing, \{\eta\}, \{\eta_1, \eta_2\}, \ldots, \{\eta_1, \eta_2, \ldots, \eta_{n-1}\}$$

correspondingly. We see that

$$|\{i < \alpha : (F^i \otimes F^{\bar{\imath}}) \cap (\cup_{\eta' \leq \eta_n} (K_\varepsilon(\eta') \otimes \{\eta'\})) = \varnothing\}| = \alpha$$

and hence

$$|\{i < \alpha : (F^i \otimes F^{\bar{\imath}}) \cap K_\varepsilon = \varnothing\}| = \alpha.$$

In particular there is $\bar{\imath}' < \alpha$ such that

$$(F^{\bar{\imath}'} \otimes F^{\bar{\imath}}) \cap K_\varepsilon = \varnothing,$$

as required. The proof that $S(X_\varepsilon) \leq \alpha^+$ for $\varepsilon = 0, 1$ is complete.

Finally we set X equal to the discrete union (i.e., the disjoint union) of the Gleason spaces of X_0 and X_1. From Corollaries 2.18(a) and 2.21 and Theorem 2.20 it then follows, using the fact that $X \times X$ contains a homeomorph of $G(X_0) \times G(X_1)$ as an open-and-closed subspace, that

$$S(X) = S(G(X_0)) + S(G(X_1)) \leq \alpha^+ \text{ and}$$
$$S(X \times X) \geq S(G(X_0) \times G(X_1)) = S(G(X_0 \times X_1)) = S(X_0 \times X_1) > \alpha^+.$$

The proof is complete.

We remark that if the extremally disconnected, compact Hausdorff space X of the statement of Theorem 7.13 is replaced by the space $\tilde{X}$ defined to be the disjoint union of X with $\beta(\alpha)$, then $\tilde{X}$ satisfies the properties required of X in Theorem 7.13 and in addition the equality

$$S(\tilde{X}) = \alpha^+;$$

indeed $\tilde{X}$ is a compact, Hausdorff, extremally disconnected space with

$$S(\tilde{X}) = S(X) + S(\beta(\alpha)) = \alpha^+,$$

and since $X \times X$ is an open-and-closed subspace of $\tilde{X} \times \tilde{X}$ we have

$$S(\tilde{X} \times \tilde{X}) \geq S(X \times X) > \alpha^+.$$

We note further that from Corollary 5.16 it follows that for a space X as in Theorem 7.13 the space $X \times X$ (even with its natural P_{α^+}-space topology) has precalibre $(2^\alpha)^+$. Thus $S(X \times X) \leq (2^\alpha)^+$ and from the hypothesis $\alpha^+ = 2^\alpha$ we have for $X, \tilde{X}$ as above the relations

$$S(X) \leq S(\tilde{X}) = \alpha^+,$$
$$S(X \times X) = S(\tilde{X} \times \tilde{X}) = (2^\alpha)^+.$$

7.14 *Remarks.* (a) The space X constructed in the proof of Theorem 7.9 has in fact the following property, stronger than property P:

every (finite) power of X has property P.

(The proof is analogous to the proof in Theorem 7.9 that X has property P.)

Problem (Kalamidas). Assume the continuum hypothesis. Is there a space X with property P such that $X \times X$ does not have property P?

(b) The spaces X_ε ($\varepsilon \in \{0, 1\}$) constructed in the proof of Theorem 7.13 have in fact the following property, stronger than the property that $S(X_\varepsilon) \leq \alpha^+$:

$$S(X_\varepsilon^n) \leq \alpha^+ \text{ for } 1 \leq n < \omega.$$

(The proof is analogous to the proof in Theorem 7.13 that $S(X_\varepsilon) \leq \alpha^+$.) It then follows from Theorem 3.27 that $S(X_\varepsilon^I) \leq \alpha^+$ for every non-empty index set I, and we have the following consequence of Theorem 7.13.

Corollary. Let $\alpha \geq \omega$ and $\alpha^+ = 2^\alpha$. There are extremally disconnected, compact Hausdorff spaces X_0, X_1 such that

$$S(X_\varepsilon^I) \leq \alpha^+ \text{ for } \varepsilon \in \{0, 1) \text{ and } I \text{ a set, and}$$
$$S(X_0 \times X_1) > \alpha^+.$$

We note in particular for $\alpha = \omega$ that the spaces X_ε are not productively c.c.c. spaces; see in this connection Problem 7.6(a) above.

7.15 *Problem* (Argyros and Negrepontis). Assume the generalized continuum hypothesis and let α, β and κ be cardinals with α and κ regular and with $\mathrm{cf}(\beta) < \kappa < \beta < \beta^+ = \alpha$.

(a) Are there spaces X and Y such that $S(X) = \alpha, S(Y) = \kappa$, and $S(X \times Y) > \alpha$? The answer is unknown even for $\mathrm{cf}(\beta) = \omega$, $\kappa = \omega^+$. We conjecture that there are such spaces. Such a space X will (from Theorem 2.2(a), with α, β, γ and δ replaced by $\alpha, \kappa, 2$ and 2, respectively) satisfy $S(X) = \alpha$ and will not have calibre α (nor even calibre $(\alpha, \kappa, 2)$), thus in a sense improving the space of Theorem 5.28.

(b) Is there a space X such that $S(X \times Y) \leq \alpha$ for every space Y with $S(Y) \leq \kappa$, but X does not have calibre $(\alpha, \kappa, 2)$? We do not even know, for $\alpha = \beta^+$ with $\mathrm{cf}(\beta) = \omega$, if there is a space X such that $S(X \times Y) \leq \alpha$ for every c.c.c. space Y, but X does not have calibre $(\alpha, \alpha, 2)$.

7.16 *Problems* (Negrepontis). Assume the generalized continuum hypothesis, and let α be a regular limit cardinal that is not weakly compact.

(a) Is there a space X such that $S(X) = \alpha$ and $S(X \times X) > \alpha$? (Jensen (1972) has defined such a space using Gödel's axiom V = L.)

(b) Is there a space X such that $S(X \times Y) = \alpha$ for every space Y such that $S(Y) \leq \alpha$, but X does not have calibre $(\alpha, \alpha, 2)$?

(c) Is there a space X such that $S(X) = \alpha$ but X does not have calibre α?

Of course a positive answer to either (a) or (b) implies a positive answer to (c).

7.17 *Problem.* If in the examples of Kunen, and of Laver and Galvin, we strengthen the topologies so that they include the usual product topology, do the resulting spaces retain the essential topological features of the original spaces?

7.18 *Remark.* We have given five examples of compact spaces with no strictly positive measure.

(1) The Galvin–Hajnal space (6.32); this has calibre ω^+.

(2) Gaifman's space (6.23); this has property K_n for $2 \le n < \omega$ and, assuming the continuum hypothesis, it does not have calibre ω^+.

(3) The examples of Argyros (6.26); for $2 \le m < \omega$ there is an Argyros space with property K_m which, assuming the continuum hypothesis, does not have property K_{m+1}.

(4) Kunen's space (7.9); assuming the continuum hypothesis, this is productively c.c.c. and does not have property (K).

(5) The Laver–Galvin space (7.13); assuming the continuum hypothesis, this is a c.c.c. space that is not productively c.c.c.

On the other hand a compact space with a strictly positive measure has, by the result of Argyros and Kalamidas (6.15), property K_n for every natural number $n \ge 2$; the Erdős space of 6.18 has a strictly positive measure and, assuming the continuum hypothesis, it does not have calibre ω^+.

7.19 *Remarks.* (a) The diagram of implications that emerges (for a space X) from the foregoing positive and negative results is as follows (p. 198). The numbers in parenthesis refer to the appropriate section of this book, the symbol (o) indicates that the implication is obvious, and square brackets indicate special set-theoretic assumptions.

(b) The diagram on page 199 summarizes those results of the preceding chapters which respond to the following question.

Assume the generalized continuum hypothesis, let α and κ be regular cardinal numbers with $\alpha \ge \omega$ and $\kappa \ge \omega^+$, and let X be a space such that $S(X) = \kappa$; does X have precalibre α?

Here the symbol N means 'no, never'; Y means 'yes, always'; and S means 'sometimes'. When S appears it is accompanied by the symbols + and − ; these indicate respectively instances in which X does and does not have precalibre α.

7.20 *Problem.* We indicate below in the Notes to this chapter that the conclusion of theorem 7.13 cannot be established in ZFC if the assumption $\alpha^+ = 2^\alpha$ is omitted: there are models of ZFC in which the product of (arbitrarily many) c.c.c. spaces is a c.c.c. space. It is fascinating nevertheless to wonder if a partial analogue of separable

(o)↓ ⇞B.3

production of separable spaces

3.7(a) 6.8↓ ⇞6.9

precalibre α, all regular $\alpha > \omega$ has a strictly positive measure

[CH]6.18

(o)↓ ⇞4.5 6.2(a)↓ ⇞6.35, 6.37

precalibre ω^+ 6.32 property (**)

(o)↓ ⇞[CH]6.23 6.2(b)↓ ⇞6.23, 6.26

property K_n, all $n \geq 2$ 6.17 property (*)

(o)↓ ⇞[CH]6.26 [CH]6.25 6.22(a)↓ ⇞?6.34

property K_m, some $m \geq 3$ (ω, ω)-chain condition

6.22(b)

(o)

property (K)

7.4(a)↓ ⇞[CH]7.9

property P

7.4(b)↓ ↑?7.6(a)

productively c.c.c.

3.3(a), 3.27↓ ⇞[CH]7.14(b)

all powers c.c.c.

(o)↓ ⇞[CH]7.13

precalibre (ω^+, ρ), all $\rho < \omega^+$ $\underset{\overrightarrow{(o)}}{\overset{7.2}{\longleftarrow}}$ c.c.c. $\underset{5.16}{\overset{\longrightarrow}{\nleftarrow}}$ precalibre α when $\omega^+ \ll \alpha, \omega^+ \ll \mathrm{cf}(\alpha)$

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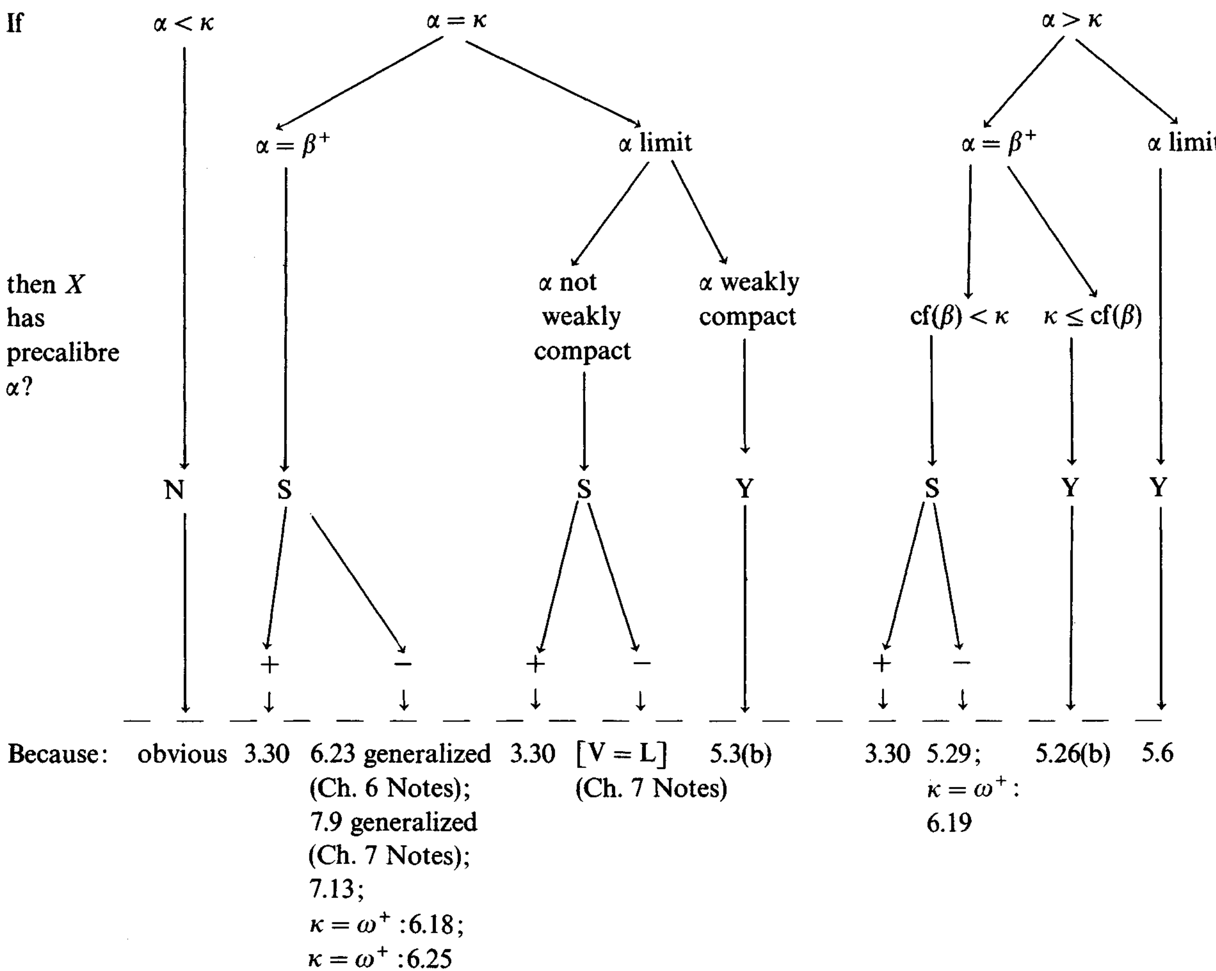
If
$\alpha < \kappa$
$\alpha = \kappa$
$\alpha > \kappa$
$\alpha = \beta^+$
α limit
$\alpha = \beta^+$
α limit
then X has precalibre α?
α not weakly compact
α weakly compact
$\mathrm{cf}(\beta) < \kappa$
$\kappa \leq \mathrm{cf}(\beta)$
N
S
S
Y
S
Y
Y
+
−
+
−
+
−
Because:
obvious
3.30
6.23 generalized (Ch. 6 Notes); 7.9 generalized (Ch. 7 Notes); 7.13; $\kappa = \omega^+$:6.18; $\kappa = \omega^+$:6.25
3.30
[V = L] (Ch. 7 Notes)
5.3(b)
3.30
5.29; $\kappa = \omega^+$: 6.19
5.26(b)
5.6

Theorem 7.13 is available in ZFC without additional special set-theoretic assumptions (in particular, without any segment of the generalized continuum hypothesis). We are thinking here of a phenomenon analogous to the following. In the theory of ultrafilters it is shown, assuming $\alpha^+ = 2^\alpha$, using a diagonal argument based on a disjoint refinement lemma not very different qualitatively from Lemma 7.11, that the set of ultrafilters uniform over α contains a large set (indeed, of cardinality 2^{2^α}) of ultrafilters minimal in the Rudin–Keisler order. This result fails in some models of the axiom system $[\mathrm{ZFC} + (\alpha^+ < 2^\alpha)]$, but in ZFC alone, using an entirely different technique based on families of large oscillation ('independent families'), the following weaker statement can still be proved: there do exist uniform ultrafilters pairwise incomparable in the Rudin–Keisler order.

Of course we can consider similar questions concerning the other examples of Chapters 6 and 7 that use the (generalized) continuum hypothesis.

Notes for Chapter 7

The principal content of Theorem 7.4, that every space with property (K) is productively c.c.c., is due to Galvin; the intermediate 'property P' used here was designed to facilitate efficient use of Theorem 7.3, an unpublished result of Kunen (1978).

Theorem 7.9, also due to Kunen, appears in Wage (1979) (Theorem 6); the proof given here, based on Lemma 7.7, is Galvin's.

The results of 7.7 and 7.8 are from Hajnal (1960).

It was Laver who first proved (unpublished) Theorem 7.13; the simpler construction given here is due to Galvin (1980). Without much additional difficulty Galvin (1980) shows that for $\alpha \geq \omega$, $\alpha^+ = 2^\alpha$ and n a natural number, there is an extremally disconnected, compact Hausdorff space X such that $S(X^n) = \alpha^+$ and $S(X^{n+1}) > \alpha^+$.

We thank Fred Galvin for several helpful, detailed letters and for unpublished, handwritten notes (34 pages, 1976).

The proofs of Theorems 6.32, 7.9 and 7.13 were given originally in the context of graph theory and the generic topology of partially ordered sets. Perhaps our (equivalent) topological presentation makes the proofs more transparent.

Extending Theorems 7.9 and 7.13, Kalamidas (1981) has proved the following result.

Theorem. Assume the generalized continuum hypothesis, and let λ be an (infinite) regular cardinal.

(a) There is a space X such that X does not have calibre $(\lambda^+,\lambda^+,2)$, and $S_\lambda(X \times Y) = \lambda^+$ for every space Y such that $S_\lambda(Y) = \lambda^+$; and

(b) there are spaces X and Y such that $S_\lambda(X) = S_\lambda(Y) = \lambda^+$, and $S_\lambda(X \times Y) > \lambda^+$.

We indicate how Martin's axiom (together with the denial of the continuum hypothesis) affects the chain conditions investigated in Chapters 6 and 7. Briefly, it trivializes or collapses the spectrum of properties between the countable chain condition and (pre-) calibre ω^+, and it has no effect on questions concerning strictly positive measures.

We give the (standard) topological equivalent of Martin's axiom. The original formulation, given by Solovay & Tennenbaum (1971), Martin & Solovay (1970) and Kunen (1968), was motivated by the forcing conditions of Cohen (1966).

Martin's axiom. If X is a compact Hausdorff space with c.c.c., $\alpha < 2^\omega$ and $\{D_\xi : \xi < \alpha\}$ is a set of dense, open subsets of X, then $\cap_{\xi<\alpha} D_\xi$ is a dense subset of X.

The basic consistency result concerning Martin's axiom is the following statement (cf. Solovay & Tennenbaum 1971).

Theorem. The axiom system [ZFC] is equi-consistent with the system [ZFC + Martin's axiom + $\omega^+ < 2^\omega$].

The effect of Martin's axiom on the countable chain condition is described by the following result.

Theorem. Assume Martin's axiom and assume $\omega^+ < 2^\omega$.

(a) If α is an infinite cardinal such that $\omega^+ \leq \mathrm{cf}(\alpha) \leq \alpha < 2^\omega$ and X is a c.c.c. space, then X has precalibre $(\alpha, \mathrm{cf}(\alpha))$.

(b) Every c.c.c. space has precalibre ω^+.

(c) The class of c.c.c. spaces is closed under the formation of products.

Proof. (a) It is by Corollary 2.22 enough to consider the case that X is the (compact, Hausdorff) space $G(X)$. If $\{U_\eta : \eta < \alpha\}$ is a set of non-empty, open subsets of X and if

$$V_\xi = \cup \{U_\eta : \xi \le \eta < \alpha\} \text{ for } \xi < \alpha$$

then from Lemma 7.1 there is $\bar{\xi} < \alpha$ such that

$$V_\xi \text{ is dense in cl } V_{\bar{\xi}} \text{ for } \bar{\xi} \le \xi < \alpha.$$

Since $\mathrm{cl}\, V_{\bar{\xi}}$ is a compact, Hausdorff c.c.c. space and $\alpha < 2^\omega$, we have from Martin's axiom that there is

$$p \in \cap \{V_\xi : \bar{\xi} \le \xi < \alpha\};$$

it is then clear that there is $A \subset \alpha$ with $|A| = \mathrm{cf}(\alpha)$ such that $\cap_{\eta \in A} U_\eta \ne \varnothing$, as required.

(b) is the case $\alpha = \omega^+$ of (a).

(c) From (b) and Theorem 7.4 it follows that (under the assumptions of the theorem) every c.c.c. space is productively c.c.c. Thus the product of finitely many c.c.c. spaces is a c.c.c. space, and the required statement (for arbitrary products) follows from (the case $\lambda = 2$, $\alpha = \omega^+$ of) Theorem 3.25 or, alternatively, from (the case $\alpha = \omega^+, \beta = \gamma = 2, \kappa = \omega, Y = X_I$ of) Theorem 3.5(a).

All essential parts of the preceding argument appear in Juhász (1970); they are attributed there to Kunen, and also to Solovay & Rowbottom.

For expository surveys of some uses of Martin's axiom, the reader may consult Martin & Solovay (1970), M. E. Rudin (1975, section IV, 1977, and Shoenfield (1975). Some specific applications and related axioms are given by Malyhin & Šapirovskiĭ (1973), Tall (1974, 1977, 1979), Juhász (1977), Herink (1977), Juhász & Weiss (1978), Kunen & Tall (1979), Wage (1979), Argyros (1981b), Zachariades (1981), and Bell (1982a). David Fremlin is currently completing a comprehensive monograph (provisionally) titled Consequences of Martin's Axiom.

In classical work K. Gődel (1938, 1940) showed that his axiom V = L (the statement that every set is constructible, that is, roughly speaking, the statement that there is in the universe no set not required by the axioms) is consistent with ZFC; further, V = L implies the generalized continuum hypothesis. Jensen (1968, 1972) derived from V = L (in fact from his 'diamond principle', a combi-

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natorial consequence of V = L) the existence of a Souslin line (that is, a non-separable *linearly ordered* space with c.c.c.), and earlier Kurepa (1950, 1952) had shown that if X is a Souslin line then $X \times X$ does not have c.c.c. Thus we have the following diagram of implications.

$$\begin{tikzcd}
\mathrm{V} = \mathrm{L} \arrow[rr, Rightarrow, "\text{Gődel}"', "\text{(1938, 1940)}"'] \arrow[d, Rightarrow, "\text{Jensen}"', "\text{(1968, 1972)}"'] & & \omega^{+} = 2^{\omega} \arrow[d, Rightarrow, "\text{Laver, Galvin}", "\text{(7.13)}"] \\
\text{there is a Souslin line} \arrow[rr, Rightarrow, "\text{Kurepa}"', "\text{(1950, 1952)}"'] & & \text{there is } X \text{ such that } S(X) = \omega^{+} < S(X \times X).
\end{tikzcd}$$

It is clear from this and from the theorem proved above that the question 'Is the product of c.c.c. spaces a c.c.c. space?' is not settled by the axioms of ZFC; like Souslin's problem, this question is answered in one way by Martin's axiom (together with the denial of the continuum hypothesis) and quite differently by Gődel's axiom V = L.

We note that Tennenbaum (1968) has proved the consistency with ZFC of the existence of a Souslin line together with the denial of the continuum hypothesis; see also Jech (1967). In a major result whose proof occupies the latter half of Devlin & Johnsbråten (1974), Jensen shows the consistency with ZFC and $\omega^{+} = 2^{\omega}$ of the statement that there is no Souslin line.

Jensen (1972, Theorem 6.2) has shown further, still assuming V = L, that if α is a strongly inaccessible cardinal that is not weakly compact, then there is a linearly ordered space X such that $S(X) = \alpha$ and $S(X \times X) > \alpha$. It is unknown whether this implication can be proved in ZFC, or even in ZFC + GCH.

We are indebted to K. Prikry for the following remarks. It has been proved independently by Silver and Lévy (cf. Jech (1978, Exercise 35.5)) that if α is a strongly inaccessible, non-measurable cardinal with an α-complete, uniform filter p for which the Souslin number of the Stone space of the quotient Boolean algebra $2^{\alpha}/p$ is equal to α, then α is not weakly compact and in fact there is a space

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X such that $S(X) = \alpha$ and $S(X \times X) > \alpha$. It has been shown by Kunen (1978) that the hypothesis of this theorem is consistent with ZFC.

For properties of the Rudin–Keisler order cited in Problem 7.20, see Comfort & Negrepontis (1974), especially Theorems 9.13 and 10.4. It was Kunen (1972) who showed that there is a set S (with $|S| = 2^{\alpha}$) of uniform ultrafilters on α, pairwise incomparable in the Rudin–Keisler order. That one may choose $|S| = 2^{2^{\alpha}}$ was shown subsequently by Shelah (cf. Shelah & Rudin 1978).

Other recent applications of the technique of large independent families can be found in Kunen (1980).

Corson-compact spaces

We consider the following classes of spaces, inspired from Banach space theory. Let X be a compact Hausdorff space.

(a) X is called an *Eberlein-compact* space if there are a Banach space B and a homeomorphic embedding of X into B, where B has the weak topology (i.e., the topology induced on B by its dual B^*).

(b) X is called a *Talagrand-compact* space if the Banach space $C(X)$ is $\mathscr{K}$-analytic in its weak topology. (A topological space is *$\mathscr{K}$-analytic* if it is the continuous image of a $K_{\sigma\delta}$-space Y, i.e., a space Y of the form

$$Y = \bigcap_{n<\omega} \bigcup_{m<\omega} K_{m,n} \quad \text{with } K_{m,n} \text{ compact.)}$$

(The introduction of Talagrand-compact spaces is due to Talagrand (1977, 1979a), and was inspired in part by Rosenthal's (1974a) beautiful example of a non-weakly compactly generated subspace of $L^1\{0,1\}^{2^{\omega}}$.)

(c) X is called a *Gul'ko-compact* space if the Banach space $C(X)$ is $\mathscr{K}$-countably determined in its weak topology. (A topological space Y is *$\mathscr{K}$-countably determined* if there is a sequence $\{A_n : n < \omega\}$ of (closed) subsets of Y such that for every $x \in Y$ there is a non-empty subset $N(x)$ of ω such that $x \in N_{n\in N(x)} A_n$, and $\cap_{n\in N(x)} A_n$ is compact.)

(d) X is a *Corson-compact* space if there are a set Γ and a homeomorphic embedding of X into the ω^+-Σ-product of $\mathbb{R}^{\Gamma}$ based at

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0, i.e., into

$$\Sigma(\mathbb{R}^\Gamma) = \{x \in \mathbb{R}^\Gamma : |\{\gamma \in \Gamma : x_\gamma \neq 0\}| \leq \omega\}.$$

We set

$$c_0(\Gamma) = \{x \in \mathbb{R}^\Gamma : \varepsilon > 0 \text{ implies } |\{\gamma \in \Gamma : |x_\gamma| \geq \varepsilon\}| < \omega\} \subset \Sigma(\mathbb{R}^\Gamma).$$

The fundamental result of Amir & Lindenstrauss (1968) states that every Eberlein-compact has a homeomorphic embedding into $c_0(\Gamma)$ for some set Γ (and hence is Corson-compact).

Talagrand (1975) proved that every Eberlein-compact is Talagrand-compact, thus establishing a conjecture of Corson (1961). It is clear that every Talagrand-compact is Gul'ko-compact; Gul'ko (1979) has proved the deep result that every Gul'ko-compact is Corson-compact (see also Vašak (1980), who independently in 1976 proved (in an implicit form) the same result). That the Gul'ko-compact spaces (as defined here) coincide with those considered by Gul'ko (1979) follows from general topological results (cf. Nagata (1972, Theorems 3, 5)).

Furthermore, there is a Talagrand-compact space that is not Eberlein-compact (Talagrand (1977, 1979a)), and there is a Corson-compact space that is not Gul'ko-compact (Alster & Pol (1980), Argyros, Mercourakis & Negrepontis (1983)).

It is an open and interesting question whether there is a Gul'ko-compact space that is not Talagrand-compact.

The survey papers of Arhangel'skiĭ (1976, 1978) and Wage (1980) provide information on Eberlein-compact and Corson-compact spaces.

The behavior of these classes of spaces with respect to the chain conditions is as follows:

(i) A c.c.c. Eberlein-compact space is separable (and hence metrizable).

(ii) A Corson-compact space with calibre $\aleph_1$ is separable (and hence metrizable). Hence, assuming MA + CH, every c.c.c. Corson-compact space is separable (and hence metrizable).

(iii) It has been proved by Argyros, Mercourakis & Negrepontis (1983), answering a problem posed by Talagrand (1977, 1979a), that every Gul'ko-compact c.c.c. space is separable (and hence metrizable). In fact, they proved that if X is a Gul'ko-compact space, then $S(X) = (w(X))^+$.

It is not known if a Gul'ko-compact (or even a Talagrand-compact) space has a dense G_δ metrizable subset consisting of G_δ-points. (This is a property enjoyed by Eberlein-compact spaces; cf. Namioka (1974) and Benyamini, Rudin & Wage (1977).) These questions are due to Talagrand (1977, 1979a, 1979b), who proved a weaker result; if answered in the positive they would establish properties stronger than the chain condition proved by Argyros and Negrepontis for Talagrand-compact spaces.

(iv) It has been observed that, assuming the continuum hypothesis, there is a non-separable Corson-compact space with a strictly positive measure. In fact, one may take the example of Haydon (1978) and use Šapirovskiĭ's (1977, 1979) result: every compact space of countable tightness can be mapped by an irreducible function into a Corson-compact space (cf. Alster & Pol (1980)).

(v) Assuming the continuum hypothesis, Argyros, Mercourakis & Negrepontis (1983) have proved:

1. There is a Corson-compact space that does not have a strictly positive measure and does not have calibre $\aleph_1$ (in fact, does not have measure-calibre $\aleph_1$), but does have property (*) and property K_n for every natural number $n \geq 2$. (This is a variation of Gaifman's construction using, as in 6.23 above, the existence of Lusin sets.)

2. There is a Corson-compact c.c.c. space X such that $X \times X$ does not have c.c.c. (cf. 7.13).

3. For every $n \geq 2$, there is a Corson-compact space K such that K has property K_n but not property K_{n+1} (cf. 7.9).

4. There is a Corson-compact space K such that K does not have Knaster's property (K) but K has property P (and hence is productively c.c.c.) (cf. 6.26).

5. The result of (iv) can be proved by similar elementary methods, i.e., without recourse to Haydon's 'sophisticated' example or Šapirovskiĭ's result, but using only e.g. the Erdős example given in 6.19 (the case $\beta = \omega$).

6. It follows from the result of Šapirovskiĭ (1979, Corollary 15), on the existence of a dense set with a point-countable base in every Corson-compact space, that there are (not necessarily compact)

c.c.c. spaces with a point-countable base with chain properties as in 1 through 5 above. (This improves, for example, van Douwen's result given in Galvin (1980) of a first-countable c.c.c. space whose square is not c.c.c.

It is an open question, due to van Douwen and Negrepontis, whether there are compact, first-countable spaces with the chain properties of 1 through 5 above.

It has been proved further by Argyros, Mercourakis & Negrepontis (1983) that if X is a Corson-compact space, then there is a one-to-one bounded linear operator $T: C(X) \to c_0(\Gamma)$ (for some set Γ). Hence for every Corson-compact space X the Banach space $C(X)$ has an equivalent strictly convex norm (and in fact, using the method of Troyanski (1971), it even has an equivalent locally uniformly convex norm). From this and the results of (v) above on chain conditions there follows a negative answer, assuming the continuum hypothesis, to the question posed earlier by Argyros and Negrepontis (see the Notes for Chapter 6): does every compact c.c.c. space X such that $C(X)$ has an equivalent strictly convex norm have a strictly positive measure?